Low-rank recurrent neural networks and their role in modeling cognition

Manuel Beiran & Francesca Mastrogiuseppe

Summary

Carefully designed behavioral tasks are a powerful tool in systems neuroscience for investigating animal cognition and its neural implementations. Theoretical neuroscientists contribute to this effort by constructing simplified mathematical models of neural circuits where structure in the synaptic connectivity among pairs of neurons gives rise to task-related computations. However, developing models that are both computationally flexible and mathematically tractable remains a challenging endeavor. Low-rank recurrent neural networks are a class of models that achieve the computational expressiveness required for common behavioral tasks while remaining mathematically tractable. A key feature of these models is that they generate low-dimensional activity, which can be parameterized in terms of a few latent variables. The dynamics of these latent variables are jointly determined by the structure of recurrent connectivity and external inputs, and can be predicted analytically using mathematical techniques collectively known as mean-field theory. Latent variables offer valuable insights into computations, as they encode task-relevant quantities such as stimuli, choices, or context, and provide a straightforward path for linking models to neural data, from which latent variables are typically extracted using dimensionality reduction techniques. Low-rank RNNs have been successfully applied to a range of systems neuroscience tasks, involving both static (e.g., context-dependent decision-making) and dynamic (e.g., integration, timing) computations, revealing how microscopic interactions between pairs of neurons within large populations can give rise to macroscopic patterns of neural activity and behavior.

Keywords: *recurrent neural networks; cognitive tasks; large-scale neural recordings; neural circuits; mean-field techniques; dimensionality reduction*

Modeling brain computations with recurrent neural networks

Everyday behavior and cognition are shaped by the electrical interactions occurring across millions of interconnected neurons in the brain. One of neuroscience's central challenges is to understand how these interactions generate large-scale patterns of activity and, ultimately, drive behavior. Mathematical models play a pivotal role in this effort, providing testbeds for formulating hypotheses, investigating mechanisms, and guiding experiments.

A mathematical framework that has been central to the modeling of brain circuits and their time-varying activity since the early days of theoretical neuroscience research is recurrent neural networks (RNNs, Fig. 1A) [Dayan and Abbott, 2001]. These models encapsulate two fundamental properties of neural circuit physiology: that single neurons process input synaptic currents non-linearly, and that these inputs originate either from the local circuit (such as surrounding neurons in a cortical column) or from external sources (such as sensory receptors or other brain areas). In the recurrent neural networks considered here, each neuron is represented by a continuous variable x_i (often referred to as *activation*), and its non-linear transform $\phi(x_i)$ (referred to as *activity*). These two variables represent, respectively, abstractions of the membrane potential and the output firing rate of each neuron, and evolve over time according to the law[1]:

[1] To simplify notation, we express time in units of the activity evolution time constant, which typically ranges from 20 to 200 ms.

$$\frac{dx_i}{dt} = -x_i + \sum_{j=1}^{N} W_{ij}\phi(x_j) + I_i. \quad (1)$$

The first term on the right-hand side, often referred to as *leak*, stems from single-neuron biophysics, and ensures that the activation decays smoothly to zero in the absence of synaptic inputs. The second term accounts for synaptic inputs generated by neighboring neurons within the circuit; their strength is proportional to the activity of neighboring neurons and scaled by the synaptic connectivity strength, W_{ij}. The third and final term encodes synaptic inputs originating from outside the circuit. Note that RNNs with slightly different dynamical equations are also possible (e.g., operating at the level of discrete action potentials, including gating variables, or obeying discrete-time evolution as in most machine learning applications).

Mathematically, RNNs are a specific type of artificial neural network. The defining feature of RNNs is that the connections between neurons form loops (i.e. bidirectional connections among pairs or groups of neurons), such that it is generally not possible to determine a single direction of information flow through the network, as in simple chain-like structure. These loops introduce a temporal dimension into the model, because they allow activity to reverberate back and forth, leading to potentially complex temporal patterns.

Recurrent neural network models have made significant contributions to the understanding of the phenomenology of neural circuits in the brain. In particular, they have been instrumental in elucidating the dynamical regime of cortical networks, which is characterized by irregular and asynchronous activity patterns [Shadlen and Newsome, 1998, van Vreeswijk and Sompolinsky, 1996, Brunel, 2000, Renart et al., 2010]. They have also generated important insights on how these activity patterns can be utilized to implement the cognitive computations that are essential for survival and adaptive behavior. Early models of computations in recurrent neural networks include Hopfield networks for the storage and retrieval of associative memories [Hopfield, 1982], continuous attractors for working memory and navigation [Ben-Yishai et al., 1995, Compte et al., 2000, Burak and Fiete, 2009], and winner-take-all attractors for decision-making [Wang, 2002].

An important application of recurrent neural networks is in the context of the cognitive tasks that animal models perform in systems neuroscience labs [Romo et al., 1999, Freedman et al., 2001, Wallis et al., 2001, Roitman and Shadlen, 2002, Churchland et al., 2012, Aronov et al., 2017, Wang et al., 2018]. The behavioral measurements in these tasks are typically paired with single-cell resolution recordings of activity across large neuronal populations, with the aim of identifying the neural substrates of specific computations. Recorded activity is often complex, displaying rich temporal profiles and high levels of heterogeneity across neurons [Churchland and Shenoy, 2007, Mante et al., 2013]. Mathematical models like recurrent neural networks can therefore assist in the interpretation of experimental data. In order to create a model for a given behavioral task, the first step is to define a surrogate computational task for the RNN, which is achieved by abstracting away non-essential aspects of the task and making assumptions about the input and output signals associated with the recorded neurons. The second step involves designing a synaptic connectivity matrix, W_{ij}, so that inputs are processed via the recurrent dynamics in a way that aligns with the target computation. This approach generates model candidate circuits that, relative to biological ones, allow for a more in-depth analysis of the relationship between synaptic structure and emergent computation.

From a theoretical perspective, designing connectivity matrices for arbitrary behavioral tasks is a challenging operation. In some cases, mathematical intuition can guide the design of connectivity matrices through simple expressions, as seen in classical models [Ben-Yishai et al., 1995, Wang,

2002]. We refer to these cases as *hand-designed* models. However, this approach becomes increasingly impractical for more complex tasks, particularly those involving high-dimensional sensory inputs or high-level cognitive functions (e.g. flexibility, planning, counterfactual reasoning). To address this challenge, neuroscientists increasingly rely on optimization algorithms, which systematically explore the connectivity matrices space to generate candidate models suited to each computational task [Mante et al. 2013, Carnevale et al. 2015, Sussillo et al. 2015, Barak, 2017, Wang et al. 2018]. We refer to these networks as *optimized* or *trained* RNNs. A variety of optimization algorithms are available for recurrent neural networks [Jaeger, 2001, Sussillo and Abbott, 2009, Pascanu et al., 2013, Kingma and Ba, 2015, Marschall et al., 2020], and implementing them has become relatively straightforward, thanks to dedicated programming packages. However, optimized recurrent neural networks come with a few major drawbacks. In contrast to hand-designed models, which are characterized by a simple, explicit, and mathematically tractable structure, optimized models operate as black boxes [Sussillo and Barak, 2013]: they can be simulated easily but are difficult to analyze. This is partly due to their inherent heterogeneity, which complicates the task of isolating signals from noise in their synaptic connectivity and activity. Another limitation is that optimization procedures explore only a limited portion of the model space, and may converge to solutions that are far from the true biological one [Das and Fiete, 2020], depending on network design and optimization scheme [Sussillo et al. 2015, Kay et al. 2024]. Furthermore, mathematical tools for analyzing these models remain sparse; existing attempts focused on linearizing the dynamics around specific activity patterns [Sussillo and Barak 2013, Driscoll et al., 2024], or fitting simpler surrogate models to the RNN for further analysis [Schaeffer et al. 2020, Langdon et al. 2025]. As a result, optimized recurrent neural networks exhibit significant limitations when used for building interpretable models of the data, or to formulate experimental predictions.

Low-rank recurrent neural networks are a specific class of recurrent neural network models that address these interpretability challenges through enhanced mathematical tractability. The article's goal is to provide an overview of low-rank RNNs as mathematical models of neural circuits, focusing on their associated mean-field analytical techniques, together with their applications to neuroscience tasks. The section “Introduction to low-rank networks” illustrates the main concepts behind these models and the associated mathematical tools. Subsequently, the section “Low-rank models for cognitive tasks” reviews some of their applications as models of cognitive tasks in systems neuroscience. The section “Low-rank models for synaptic plasticity and learning” finally discusses the contributions of these models to the problem of learning in biological and artificial neural networks.

Most of the applications discussed address mammalian model organisms and cortical brain regions; however, the framework is agnostic to the specific modeled biological system, and can be applied to other species and brain regions as well.

Introduction to low-rank networks

Relevance for modeling cognition

A crucial element to specify in recurrent neural network models of cognitive tasks is the synaptic connectivity matrix, W_{ij}. Low-rank RNNs are a class of recurrent network models in which the synaptic connectivity is restricted to a simple form: a matrix of rank R, where R is a small integer.

Mathematically, the rank of a matrix is defined as the number of non-zero singular values. Any rank-R matrix can be expressed as the sum of R rank-one terms. Thus, a low-rank synaptic connectivity matrix can be written as

$$W_{ij} = \sum_{r=1}^{R} m_i^r n_j^r. \quad (2)$$

Each rank-one structure in the sum above, indexed by r, is defined by the outer product of two vectors, $\mathbf{m}^r$ and $\mathbf{n}^r$, termed connectivity patterns[2] (Fig. 1B). Intuitively, a low-rank matrix is a particularly simple structure because it can be fully described by only two vectors per rank, which significantly reduces the number of parameters in the model. While a generic synaptic connectivity in a network of N neurons has N^2 parameters – corresponding to the number of entries in the matrix, which grows quadratically with the number of neurons – low-rank synaptic connectivity has $2RN$ parameters, which only grows linearly with the network size. Another way to conceptualize a low-rank matrix is as one in which many rows are proportional to each other (and therefore also many columns). For example, in the case of rank-one connectivity, all rows are proportional to vector $\mathbf{n}^\top$, with factors of proportionality given by m_i.

The main rationale for considering low-rank RNNs is that, despite their simplicity, they possess sufficient computational expressiveness to model systems neuroscience tasks. This is exemplified by the observation that many classical RNN models of neural computations are characterized by connectivity matrices that are exactly or approximately low-rank. One such example is Hopfield networks for associative memories (Fig. 1C) [Hopfield, 1982]. In their classical implementation, the memorization of an activity pattern $\mathbf{p}$ is achieved by updating the synaptic connectivity matrix according to the Hebbian rule: connectivity between the post- and pre-synaptic neurons, W_{ij}, is strengthened or weakened by the term $p_i p_j$ where p_i and p_j represent activity of the post- and pre-synaptic neurons within that pattern. By comparing this update with Eq. 2, it becomes clear that this constitutes a rank-one update of the connectivity, where both connectivity patterns $\mathbf{m}$ and $\mathbf{n}$ correspond to the memorized pattern $\mathbf{p}$ (Fig. 1D).

Another example of classical models that can take a low-rank form is continuous attractor networks, which are capable of storing a variable in working memory: ring attractor networks for angular variables, and line attractor networks for non-periodic ones. Ring attractors have been introduced to model various phenomena, including orientation tuning in the primary cortex [Ben-Yishai et al., 1995, Tsodyks and Sejnowski, 1995, Goldberg et al., 2004], the head-direction system [Zhang, 1996, Redish et al., 1996], and working memory in the prefrontal cortex [Compte et al., 2000]. In these models, each neuron is assigned a preferred angle θ_i ranging from 0 to 2π, and the synaptic connectivity between any two neurons depends, on average, only on the difference between their preferred angles. For instance, following Ben-Yishai et al. [1995], the synaptic connectivity is given by $W_{ij} = w\cos(\theta_i - \theta_j)$, in addition to homogeneous inhibition. Notably, the connectivity can be expressed as a rank-two matrix: $W_{ij} = w(\cos\theta_i \cos\theta_j + \sin\theta_i \sin\theta_j)$. As in Hopfield networks, connectivity patterns $\mathbf{m}^r$ and $\mathbf{n}^r$ correspond to the same pattern: $\mathbf{m}^1 = \mathbf{n}^1 = \cos\boldsymbol{\theta}$ and $\mathbf{m}^2 = \mathbf{n}^2 = \sin\boldsymbol{\theta}$ (Fig. 1F). Therefore, the connectivity matrix is symmetric. Line attractors were instead introduced to model the oculomotor neural integrator, a distributed network in the hindbrain of vertebrates which integrates the velocity commands to the eyes and outputs the eye position [Aksay et al., 2001]. Decades later, those models have been associated with the encoding of affective states in the hypothalamus [Nair et al., 2023, Vinograd et al., 2024]. As observed in Seung [1996], approximate line attractors can be implemented through a rank-one synaptic connectivity matrix generating positive feedback.

[2] One standard method for decomposing a matrix in this form is through singular value decomposition. In that case, patterns $\mathbf{m}^r$ and $\mathbf{n}^r$ are proportional to the left and right singular vectors associated with the r-th nonzero singular value.

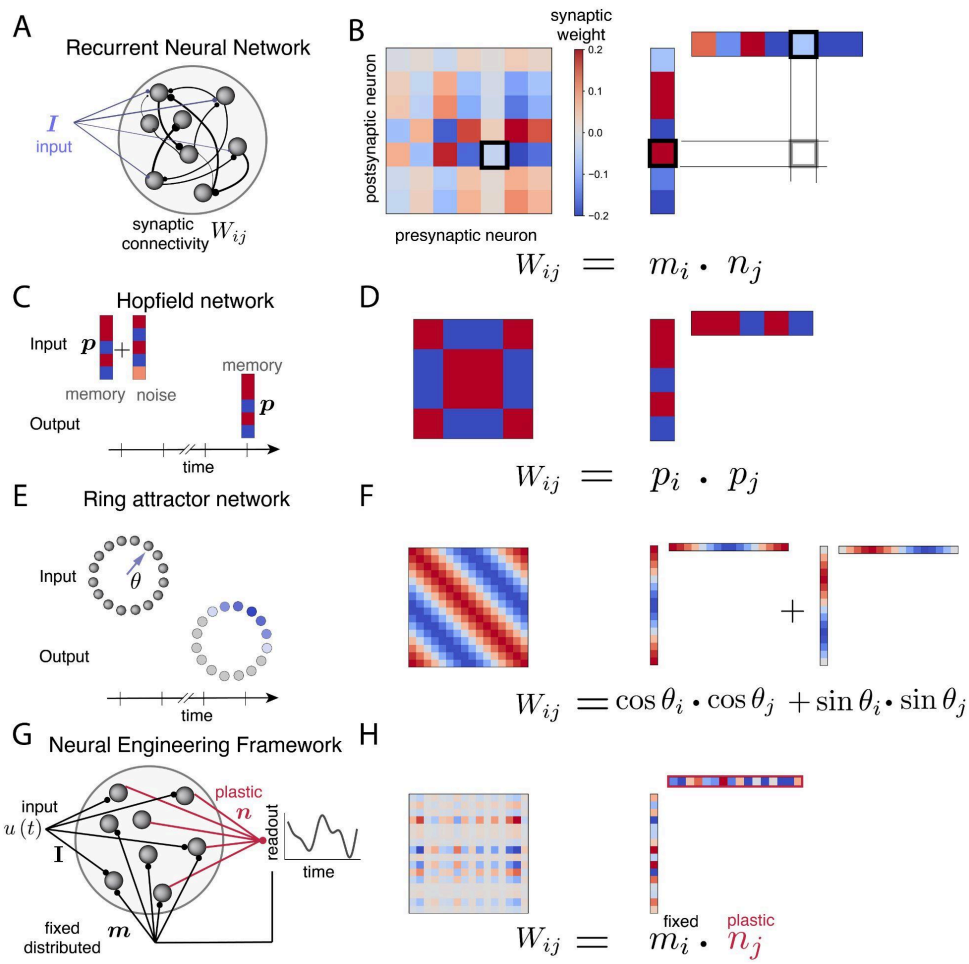

Figure 1: Examples of recurrent neural networks with low-rank connectivity structures.

A Recurrent neural network architecture. The time-varying activity of the neurons in the network is determined by the external input (violet) and the synaptic inputs currents that are generated by recurrent connections (black).

B Example of a rank-one synaptic connectivity matrix. The matrix is the outer product of the connectivity patterns **m** and **n**.

C Hopfield networks for associative memory. The activity of neurons in the network persistently produces a stored pattern, **p**, after a corrupted version of the pattern is presented as a transient input.

D A Hopfield network that only stores one pattern has rank-one connectivity. The connectivity patterns are identical to the memorized pattern, and their entries are binary.

E Neurons in a ring attractor network are sorted based on their preferred angle **θ**. After a brief input in which a target angle is presented, the network persistently generates a response around that angle.

F Simplified rank-two connectivity for the ring attractor in Ben-Yishai et al. [1995]. (Note that the original model also includes an offset).

G Simplified architecture of the Neural Engineering Framework (NEF) for learning a one-dimensional dynamical system, such as an integrator, based on Eliasmith and Anderson [2004]. A rank-one loop, where one of the connectivity patterns is plastic, is learned via linear regression. The other connectivity pattern is fixed, and so is the input pattern.

H Illustration of a NEF connectivity matrix for implementing a one-dimensional dynamical system. The resulting connectivity matrix is rank-one.

A third example of low-rank connectivity can be found in spike-coding neural networks. In this framework, the need for optimal, efficient encoding of a time-varying input signal also results in a rank-one connectivity matrix. In this case, the connectivity patterns **m** and **n** are built from the vectors along which the signal is fed into and decoded from the network [Boerlin et al., 2013, Mancoo et al., 2020, Podlaski and Machens, 2024]. It is important to note that all the models discussed thus far share a common characteristic: they perform computations on a limited number of cognitive variables (one pattern stored in a Hopfield network, one variable represented in a continuous attractor, and one temporal signal encoded in a spike-coding network).

Low-rank connectivities are not only characteristic of classical models of cognitive computations; they are also found in frameworks for the numerical optimization of RNNs. Machine learning algorithms for recurrent architectures have been available since the late 1980s [Rumelhart et al. 1986, Williams and Zipser, 1989]; however, they were technically challenging to implement at the time. Decades later, some of the first optimization algorithms for RNNs to gain popularity in neuroscience involved synaptic connectivity updates in a low-rank form. Two well-established examples are the Neural Engineering Framework (NEF) [Eliasmith and Anderson, 2004] and FORCE learning [Sussillo and Abbott, 2009]. In both algorithms, the learning rule modifies the entries of the connectivity pattern **n**, while **m** remains fixed (Fig. 1G). While NEF is an offline method that results in an exactly low-rank connectivity, FORCE learning operates online, correcting errors in real-time as neural dynamics unfold, and is based on adding a low-rank term to an otherwise random, high-rank connectivity.

In summary, low-rank synaptic connectivity matrices are widespread in recurrent network models, encompassing both classical and optimized ones. The upcoming subsection “Low-rank RNNs generate low-dimensional activity” explains why this type of connectivity represents a favorable choice: it allows for mathematical analysis, thereby enhancing model interpretability. This can be understood by examining the relationship between network activity and connectivity through a geometric lens. The subsection “Computing latent variables via mean-field theory” further discusses how this relationship can be rigorously analyzed using a set of mathematical tools collectively known as mean-field theory.

Low-rank RNNs generate low-dimensional activity

What type of neural activity do low-rank RNNs generate? To address this question, it is useful to rewrite their temporal evolution (Eq. 1) in matrix notation. By denoting the network-wide vector of activations as $\mathbf{x} = (x_1,...,x_N)^\top$, the dynamics read

$$\frac{d\mathbf{x}}{dt} = -\mathbf{x} + \sum_{r=1}^{R} \mathbf{m}^r \mathbf{n}^{r\top} \phi(\mathbf{x}) + \mathbf{I}. \tag{3}$$

On the right-hand side, the expressions $\mathbf{n}^{r\top}\phi(\mathbf{x})$ represent dot products between two vectors, resulting in scalar values. These scalars can by denoted by κ^r, such that:

$$\frac{d\mathbf{x}}{dt} = -\mathbf{x} + \sum_{r=1}^{R} \mathbf{m}^r \kappa^r + \mathbf{I}. \tag{4}$$

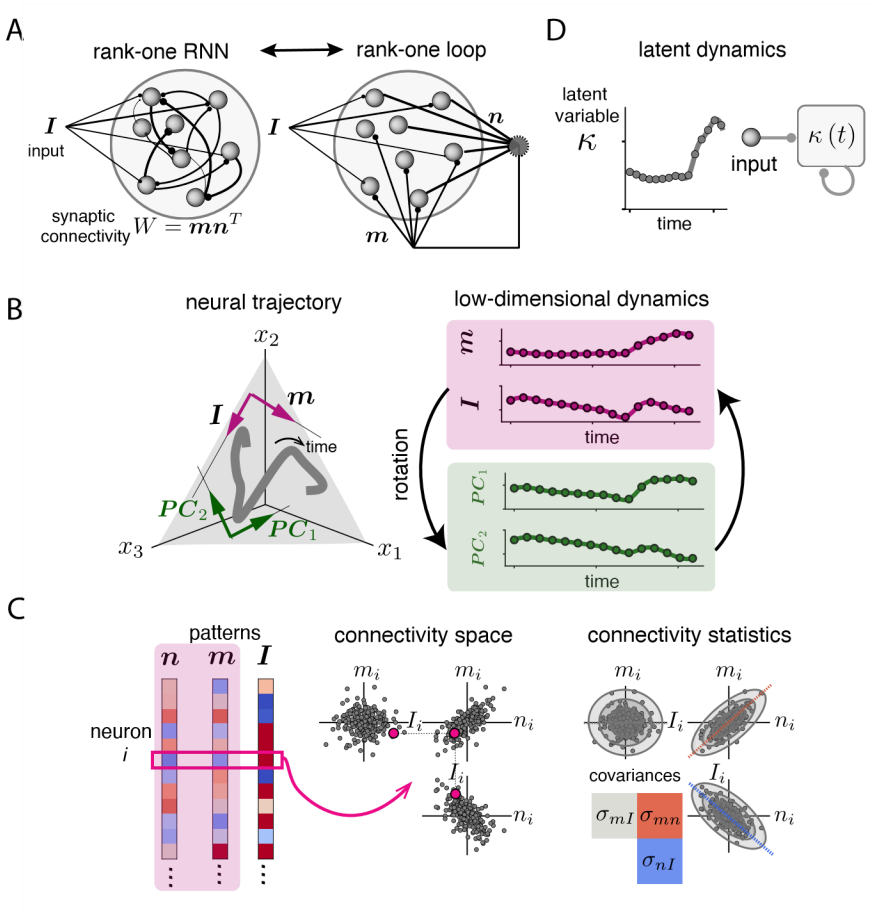

Figure 2: Characterization of low-dimensional activity in recurrent neural networks with low-rank connectivity structures.

A A recurrent neural network with rank-one connectivity (left) functions effectively as a group of neurons interacting through a connectivity loop (right): a linear unit (dashed edge) reads out activity along pattern **n**, and sends this signal back to the group of neurons along pattern **m**. A higher-rank network would be characterized by multiple linear readout units.

B Left: high-dimensional neural space, where each axis corresponds to the activation of one neuron in the network. The activity of a low-rank network is constrained to the linear subspace (shaded plane) spanned by the patterns **I** and **m**. The principal components of the activation (green arrows) identify the same subspace. Right: time-varying projection of the activation **x** along patterns **I** and **m** (top) and along the first two principal components (bottom). Both sets of projections fully describe the dynamics of the rank-one network, because the two sets of vectors are equivalent up to a rotation.

C Left: in a rank-one network, parameters are specified by the connectivity patterns **m** and **n** and the input pattern **I**. These are *N*-dimensional vectors. The *i*-th entries of these vectors (magenta rectangle) correspond to parameters associated with neuron *i*. Middle: parameters associated with each neuron can be visualized in a connectivity space where the axes represent the entries of the connectivity and input patterns, and each dot corresponds to one neuron. Right: the distribution obtained by plotting parameters of all neurons in the connectivity space can often be approximated by a smooth probability function. In this illustration, the parameters are well approximated by a single standard multivariate Gaussian, whose key parameters are the covariances (bottom left corner) [Mastrogiuseppe and Ostojic, 2018]. The ellipses indicate regions with cumulative probability equal to 1, 2, and 3 standard deviations.

D Left: activation in a rank-one network can be summarized by a latent variable κ. The stationary and time-varying behavior of κ can be described by using mean-field theory. Right: schematic representation of the

temporal evolution of the latent variable. Arrows denote dependencies. At each time point, changes in the latent variable κ depend on the external input and the latent variable itself. Those dependencies are in general non-linear. The precise functional form of the latent variable dynamics is determined by the connectivity and input statistics.

This equation and the activation variables it generates can be analyzed from a geometrical perspective, by visualizing the activation vector $\mathbf{x}$ in the high-dimensional neural space where each axis represents the activation of a single neuron (Fig. 2B, left). As time progresses, $\mathbf{x}$ traces a continuous trajectory through this space, reflecting the evolution of network activity. Crucially, Eq. 4 imposes significant constraints on this trajectory. In particular, at every point in time, the activation trajectory cannot freely vary in all directions; instead, it is confined to vary within a linear subspace defined by a limited number of vectors: the connectivity patterns $\mathbf{m}^r$, and the external input $\mathbf{I}$. This results in an activation trajectory that lies within this linear subspace[3] [Mastrogiuseppe and Ostojic, 2018].

We can confirm that the activation trajectory indeed follows these geometrical constraints by applying dimensionality reduction algorithms to the model-generated data (Fig. 2B). Dimensionality reduction is a statistical approach that provides tools to visualize high-dimensional data – such as activity from large neural populations – by identifying the prominent directions within this high-dimensional space that capture the majority of the data's variation [Cunningham and Yu, 2014]. In particular, linear dimensionality reduction techniques return a linear embedding for the data, and therefore quantify their *embedding* dimensionality [Jazayeri and Ostojic, 2021]. One example of these tools is Principal Component Analysis (PCA). When applying PCA to the activation trajectories generated by a low-rank RNN, one finds that the data's variation can be well captured by a restricted number of directions. These are correlated with the connectivity patterns $\mathbf{m}^r$ and the external input $\mathbf{I}$ (Fig. 2B), and span the same linear subspace. Therefore, the dimensionality of the activation trajectories generated by low-rank RNNs is low[4].

There exists an alternative, although equivalent, way to visualize the link between low-rank connectivity and low-dimensional activity. So far, the explanation has focused on the second term of the right hand sides of Eqs. 3 and 4: the synaptic input currents generated from recurrent interactions across neurons. Eq. 4 indicates that, in a low-rank RNN, these interactions are mediated by a small number of scalars, κ^r. This contrasts with a generic RNN, where interactions are mediated by the network-wide activity, $\phi(\mathbf{x})$, which can, in principle, be high-dimensional. This structure can be depicted as interactions across all pairs of neuron being routed through a low-dimensional bottleneck parameterized by the scalar signals κ^r (Fig. 2A). The bottleneck can be viewed as an abstract entity governing network dynamics, or as a true biological entity (e.g. an additional set of neurons, whose activity is related to the scalar signals, as in Logiaco et al. [2021], Kao et al. [2021], DePasquale et al. [2023]).

[3] Strictly speaking, Eq. 4 indicates that the activation at time t can also vary along the direction defined by $\mathbf{x}(t)$ itself. Consequently, if $\mathbf{x}(t)$ does not lie within the $\mathbf{m}$-$\mathbf{I}$ plane – for instance, because the activation is initialized with random values at $\mathbf{x}_0$ – the dynamics will include an additional component along that direction. This additional component, however, will rapidly decay to zero, as the only directions that can be associated with large eigenvalues (and, therefore, slow dynamics) in low-rank RNNs are the connectivity patterns. Alternatively, out-of-plane initial conditions can be interpreted as arising from a pulse-like external input along $\mathbf{x}_0$. In this perspective, the linear subspace spanned by the activation is transiently three-dimensional, generated by one connectivity and two distinct input patterns.

[4] More precisely, the number of dimensions spanned by the activation trajectories in a low-rank RNN is at most the sum of the network rank plus the number of input patterns.

All in all, imposing a low rank on the synaptic connectivity matrix W effectively enforces low dimensionality on the network's activation state[5]. This naturally raises the question of whether low-dimensional activity represents an acceptable model byproduct when the goal is to capture large-scale experimental recordings during behavioral tasks. Several studies have estimated the dimensionality of neural activity in large neuronal populations, employing different behavioral setups, recording from various brain areas, utilizing diverse recording techniques, and applying different dimensionality metrics [Gao and Ganguli, 2015, Cueva et al., 2020, Litwin-Kumar et al., 2017, Stringer et al., 2019, Lanore et al., 2021, Manley et al., 2024]. The findings from this diverse body of work are not uniform and suggest (perhaps unsurprisingly) that the dimensionality of neural activity depends on a combination of these factors altogether. However, there appears to be a consensus that, at least in the context of behavioral tasks involving slow dynamics [Cueva et al., 2020, Wang et al., 2018], with a limited number of input variables [Mante et al., 2013, Chaisangmongkon et al., 2017], and, particularly, in recordings from cortical areas associated with high-level cognitive processes [Brincat et al., 2018], the dimensionality of neural activity is indeed low (i.e., it is typically an order of magnitude smaller than the number of recorded neurons). In that context, it is therefore reasonable to consider low-rank RNNs as a suitable tool for modeling and analyzing experimental data.

Computing latent variables via mean-field theory

Low-dimensional activity offers a key advantage: it is inherently simpler to visualize and characterize. This is because it can be fully specified by a small set of scalars, denoted here by κ^r, which define the coordinates of the activation trajectory within its low-dimensional subspace (Fig. 2B). Drawing from terminology commonly used in state-space modeling and dimensionality reduction [Cunningham and Yu, 2014], these scalars are referred to as *latent variables*.

A key strength of low-rank RNNs is that, in many scenarios, the behavior of latent variables – and, consequently, the behavior of network-wide activity – can be predicted mathematically using a set of analytical techniques collectively known as *mean-field* approaches [Sompolinsky et al., 1988, Moritz and Dahmen, 2020, La Camera, 2022]. Originating from statistical physics and complex networks, these approaches often assume that the RNN contains a relatively large number of neurons (often, a few hundred units, as seen in classical and optimized RNNs, are sufficient). The core idea of mean-field theory is that one can provide a compact description of network-wide activity by abstracting away details about individual neurons' activity and connectivity. This is achieved by treating single neurons as samples drawn from a smooth probability distribution, which can be computed given knowledge of the RNN's connectivity and external inputs.

To illustrate how mean-field theory is applied to low-rank RNNs, it is sufficient to consider the straightforward scenario of computing the stationary states of the dynamics, which involves setting the right-hand side of Eq. 4 to zero [Strogatz, 2000]. The stationary activation, $\mathbf{x}_*$, obeys

$$\mathbf{x}_* = \sum_{r=1}^{R} \mathbf{m}^r \kappa_*^r + \mathbf{I} \tag{5}$$

where we have defined $\kappa_*^r = \mathbf{n}^{r\top}\phi(\mathbf{x}_*)$. As expected, the activation lies on the hyperplane spanned by connectivity patterns and external inputs; to exactly locate them, one must compute the latent

[5] Note that the non-linearly transformed activity $\phi(\mathbf{x})$ will have, in principle, higher dimensionality. The difference in dimensionality is small when most neurons operate in semi-linear regions of the activation function ϕ.

variables κ_*^r [6]. In order to do so, the latent variables can be re-expressed as sums running over all neurons in the network:

$$\kappa_*^r = \sum_{j=1}^{N} n_j^r \phi\left(x_{*j}\right). \tag{6}$$

In line with standard mean-field approaches, it is possible to interpret these sums as averages: $\kappa_*^r \propto \langle\, n_j^{r\,\top} \phi(x_{*j}) \,\rangle_j$, where $\propto$ indicates equality up to a scaling factor and angular brackets denote averages over neurons. In principle, the probability distribution specifying the joint behavior of n_j^r and x_{*j} across neurons would enable the computation of these averages using standard techniques from probability theory and statistics. However, at this point, how x_{*j} behaves has not been specified yet. Luckily, from Eq. 5, x_{*j} can be re-expressed in terms of the latent variables κ_*^r, the connectivity patterns and external inputs, as

$$\kappa_*^r = \sum_{j=1}^{N} n_j^r \phi\left(\sum_{r=1}^{R} m_j^r \kappa_*^r + I_j\right). \tag{7}$$

The averages in Eq. 7 can be computed using a probability distribution, which does not involve the activation x_{*j}, but only the value of connectivity pattern entries m_j^r and n_j^r for all ranks, as well as the external input entries I_j (Fig. 2C left and middle). This probability distribution is a relatively simple object: for a rank-one RNN, it is a function of three variables whose shape can be controlled by a handful of parameters (Fig. 2C right); for a rank-two RNN, it would be a function of five variables, and so on. When a low-rank RNN is hand-designed, this probability distribution is known a priori. On the other hand, when the low-rank RNN arises from a fitting or optimization procedure, the probability distribution can be empirically estimated from the entries of the connectivity and external input patterns. Once this probability distribution is established, statistical averages can be finally calculated via standard analytical and numerical methods, resulting in a self-consistent equation for the latent variables κ_*^r [Mastrogiuseppe and Ostojic, 2018].

In principle, low-rank RNNs can be generated from any choice of probability distribution[7]. In practice, however, work on low-rank RNNs has focused on probability distributions that take particularly simple forms [Hopfield, 1982, Ben-Yishai et al., 1995, Mastrogiuseppe and Ostojic, 2018, Pereira and Brunel, 2018, Beiran et al., 2021, Schuessler et al., 2020a, Susman et al., 2021]. This focus ensures that the mathematical expressions for averages can be easily solved. Specifically, Mastrogiuseppe and Ostojic [2018] examined connectivity and input patterns generated from a multivariate Gaussian distribution, where the key parameters to be specified correspond to the covariances (Fig. 2C right). Later research [Beiran et al., 2021, Dubreuil et al., 2022] considered instead mixtures of multivariate Gaussians. This choice significantly enhances the expressiveness of the models by expanding the range of dynamics they can produce and the computations they can implement, while still maintaining manageable mathematical analysis. An example probability distribution generated as a mixture of two multivariate Gaussians is shown in Fig. 3A. As parameters for each neuron are drawn from one of the two Gaussian distributions (green or purple, Fig. 3A), the

[6] The entries of **n** are assumed to have magnitude that scales with the network size as $1/N$, while entries of **m** have fixed magnitude. Under this assumption, the magnitude of latent variables does not depend on the network size. This choice is consistent with the connectivity scaling used in Hopfield [1982] and Mastrogiuseppe and Ostojic [2018]. But different connectivity scalings are also possible [Landau and Sompolinsky, 2018].

[7] Note that, based on this reasoning, different low-rank RNNs will exhibit identical latent variables and, more generally, identical low-dimensional dynamics whenever they share the same generating probability distribution. This holds true as long as the networks are sufficiently large to ensure that the distribution is adequately sampled, i.e., there are enough neurons for each possible combination of connectivity and external input values.

RNN can be conceptualized as being organized into two populations, each associated with different connectivity and input statistics (Fig. 3B).

Mean-field Eqs. 5–7 refer to stationary patterns of network activity, also known as fixed points. However, like any RNN, low-rank models can also generate more complex activity patterns, including continuous manifolds of fixed points [Beiran et al., 2021], oscillations [Mastrogiuseppe and Ostojic, 2018], chaotic dynamics [Pereira and Brunel, 2018], and transient amplification responses due to non-normality [Bondanelli and Ostojic, 2020]. In these scenarios, dynamics in the RNN can still be studied through its latent variables, which exhibit non-trivial temporal evolution (Fig. 2D). Developing mathematical tools for this full repertoire of dynamical patterns is essential, as different activity types may underlie the implementation of distinct components and time epochs of computational tasks. Available methods for describing non-stationary activity draw upon tools from mean-field theory, conceptually similar to those used for stationary solutions. They also incorporate techniques from random matrix theory [Girko, 1985, Rajan and Abbott, 2006, Tao, 2013], dynamical mean-field [Sompolinsky et al., 1988, Pereira and Brunel, 2018, Mastrogiuseppe and Ostojic, 2018, Landau and Sompolinsky, 2018], and linear algebra [Bondanelli and Ostojic, 2020, Beiran et al., 2021].

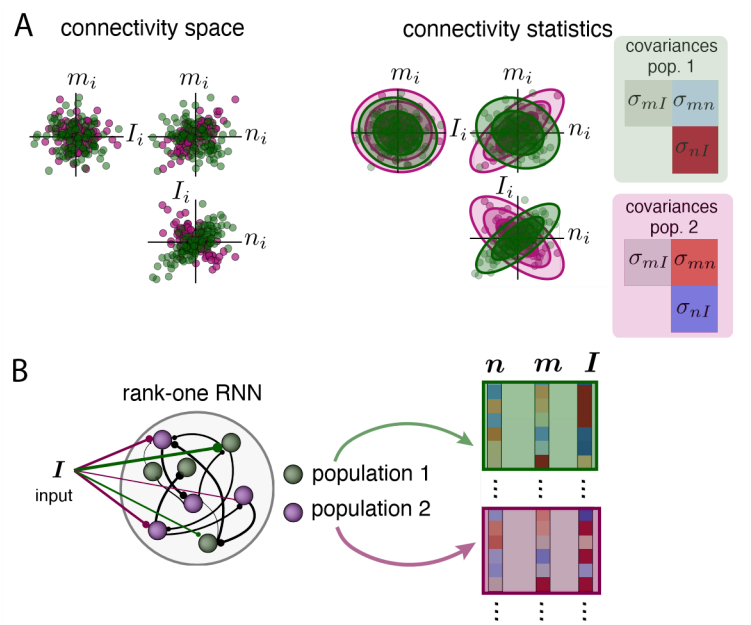

Figure 3: Low-rank recurrent neural networks with connectivity drawn from a mixture of Gaussians distributions. Each Gaussian can be conceptualized as representing a sub-population of neurons with homogeneous connectivity and input statistics [Dubreuil et al., 2022, Beiran et al., 2023].

A Left: visualization of the connectivity space for a rank-one network generated from a mixture of two Gaussian distributions (green and purple). Connectivity and input parameters for each neuron are drawn accordingly to one of the two distributions. Right: probability distribution from which the sample in A is drawn. Note that the two Gaussians are parameterized by different covariances.

B The rank-one network in A can be conceptualized as a RNN with two populations (green and purple). Neurons in the two populations are associated with different Gaussian distributions.

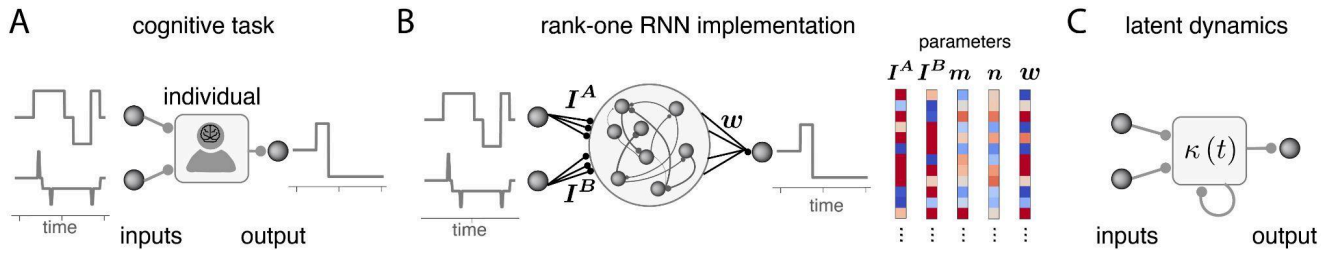

Figure 4: Cognitive tasks and their implementation in low-rank networks.

A A cognitive task is defined by a set of inputs (e.g. stimuli) and a set of desired outputs (e.g. actions), both defined by the experimentalist.

B Left: rank-one RNN implementation of a cognitive task. In this illustration, there are two time-dependent input variables, fed to the network along patterns $\mathbf{I}^A$ and $\mathbf{I}^B$. The network output is given by the linear projection of activity $\mathbf{w}^\top\phi(\mathbf{x})$, where $\mathbf{w}$ is the readout pattern. Right: optimizable network parameters, corresponding to the entries of five patterns: two input patterns, two connectivity patterns and the readout.

C Schematic representation of the non-linear temporal evolution governing the latent variable in the trained rank-one RNN. Arrows denote dependencies between variables, with the output variable representing a transformation of the latent variable at each time point. By adjusting the connectivity and input parameters and their statistics, the optimization process shapes the functional form of the latent variable dynamics.

Low-rank models of cognitive tasks

Building low-rank models via numerical optimization

How can one employ the mathematical framework of low-rank recurrent neural networks to construct interpretable circuit models for systems neuroscience tasks? A cognitive task is in general defined by a set of inputs (e.g., sensory stimuli, contextual information, etc), which may vary over time, and a rule to associate those inputs to particular actions (Fig. 4A). In a laboratory setting, the inputs and the mapping from inputs to desired outputs are set by the experimentalist. These mappings can be arbitrarily complex – for instance, many cognitive tasks include a delay period between stimulus presentations and motor actions, requiring working memory and motor planning. Just as the brain of the participating animal stands in between the task inputs and the motor responses, recurrent network models of behavioral tasks receive inputs and generate task-constrained output responses. The output of an RNN is usually defined as the projection of the neural activity along a readout pattern $\mathbf{w}$; for tasks with multiple outputs, each output is associated with a distinct readout pattern[8].

RNN models of computational tasks can be derived either through design or optimization; the latter strategy is useful when there are no a priori hypotheses about which dynamical landscape can effectively implement a given task [Barak, 2017]. In this process, some or all of the network parameters are adjusted until the network's output aligns with the desired task output. This principle also applies to low-rank RNNs. The most straightforward approach to optimize a low-rank RNN is to

[8] When modeling behavioral tasks with RNNs, it is necessary to make explicit assumptions about the task's input and output signals, often with several valid options. Brain-area-specific information can help narrow these choices, but a common approach is to encode only the minimal required information in the inputs and enforce the network's recurrent dynamics to generate the additional required signals.

use standard gradient-based optimization algorithms, such as backpropagation through time, on models where connectivity matrices are explicitly parameterized in a rank-R form (Eq. 2) [Dubreuil et al., 2022]. In this case, the optimization algorithm operates directly on the entries of the connectivity patterns $\mathbf{m}^r$ and $\mathbf{n}^r$. The rank R serves as a hyperparameter; the most parsimonious models will be those that can be successfully trained on the task with the minimal rank R. For example, in a rank-one network receiving two different input variables, the optimized parameters are the input patterns $\mathbf{I}^A$ and $\mathbf{I}^B$, the connectivity patterns $\mathbf{m}$ and $\mathbf{n}$, and the readout pattern $\mathbf{w}$ (Fig. 4B). Interestingly, networks constrained to have different ranks may return different strategies to solve the same task (for example, Beiran et al. [2023] found that networks with low-rank constraint had better generalization properties than full-rank networks).

Once the RNN is optimized, its activity can be analyzed using theoretical tools for low-rank networks. Specifically, the dynamics of the high-dimensional activation can be reduced to latent variables, which encode task-relevant signals and therefore are related to the network's output (Fig. 4C). Additionally, mean-field theory can be applied, provided that the joint probability distribution governing the connectivity and input entries (Fig. 2C) is known. The exact mathematical shape of this distribution in optimized low-rank RNNs is unknown, and likely depends on various factors (such as the form of the activation function and the distribution of parameters at initialization). In practice, however, for sufficiently large RNNs, one can often use a simple functional form to fit this empirical distribution and achieve good performance. Dubreuil et al. [2022] found that probability distributions that are a mixture of a few (1-3) multivariate Gaussian distributions effectively fit low-rank models optimized on a variety of common behavioral tasks.

As an alternative approach to optimizing fixed-rank RNNs, one can optimize RNNs in an unconstrained manner (for example, by employing gradient-based optimization directly on each entry of the connectivity matrix W_{ij}) and subsequently attempt to distill a low-rank approximation from the resulting model. At least in principle, this method is not guaranteed to succeed, given that the optimization process may converge to solutions that are not well represented by a low-rank model. However, it has been shown that the connectivity updates that emerge from standard gradient-based optimization algorithms are often low-rank [Schuessler et al., 2020b]. This results in a final connectivity that is the sum of the initial connectivity (which is typically task-agnostic and full-rank) and a low-rank matrix (as in Fig. 1H). Mean-field tools for analyzing RNNs with this form of composed connectivity have been developed [Schuessler et al., 2020a,b, Susman et al., 2021]. Yet, these techniques are mathematically and computationally more demanding, due to the need to account not only for the statistical dependencies among the entries of the connectivity and input patterns, but also for their correlations with the random entries of the full-rank initialization matrix.

Combining mean-field and optimization tools, one can construct RNN models of cognitive tasks that can be both qualitatively and quantitatively understood. To illustrate this, the two upcoming subsections highlight low-rank models proposed for two different cognitive tasks: a context-dependent decision-making task, inspired by Mante et al. [2013], which requires a flexible mapping from stimulus to action based on context, and an interval timing task, inspired by Wang et al. [2018], which demands precise timing of the output action. A specific focus is placed on illustrating the connectivity structures that these tasks impose on recurrent networks, and their relationship to the computations performed.

Context-dependent decision making task

We first focus on a contextual decision-making task based on the study by Mante et al. [2013] (Fig. 5A). In this task, there are two sensory stimuli variables (A and B) and two contextual variables (A and B). In each trial, the two sensory variables fluctuate randomly over time around different average

levels. Additionally, one of the two context variables is active. When the context A variable is active, the network must internally process the sensory variable A, to determine whether its average value over time is positive or negative, and ignore sensory variable B. Conversely, when the context B variable is on, the network must ignore the sensory variable A, and report about the sensory variable B. This context-dependent decision rule requires flexibility in how the sensory space is mapped to the action on each trial (Fig. 5A right).

Dubreuil et al. [2022] trained recurrent neural networks with rank-one connectivity to solve this task, and then used the low-rank theoretical framework to link the structural properties of the network with the implemented computation. The optimized connectivity and input parameters of the network displayed a highly structured organization (Fig. 5B). This structure was effectively captured by a mixture of two Gaussian distributions (green and purple, Fig. 5B), whereas fitting a single Gaussian produced a network incapable of producing context-dependent outputs. This finding suggests that one can conceptualize the network's function as relying on two distinct neuronal populations, each receiving statistically different stimulus and context inputs, and governed by different recurrent connectivity rules.

A key advantage of low-rank RNNs is that their activity can be distilled into a few scalar signals – the latent variables – which provide crucial insights into the task implementation, and whose dynamics can be predicted through mean-field theory. For the rank-one network in Fig. 5B, Dubreuil et al. [2022] found that the latent variable κ is directly related to the network output, representing integrated evidence towards one of the two possible output choices, and converging to one of two possible output states by the end of the trial (Fig. 5C). The effective dynamics of κ are driven by the two sensory variables A and B; these dynamics are slow, allowing for the integration of sensory variables over the course of the trial.

To meet task requirements, only one sensory input is integrated per trial, depending on context. This is achieved through context variables, which act as gain modulators for the sensory inputs driving the latent variable (Fig. 5C, black). In context A trials, contextual input A reduces the gain of sensory input B, and vice versa. This gating effect, which effectively reshapes sensory inputs in two possible configurations based on context, could be traced back to the presence of two distinct populations in the network. Specifically, the authors showed that gating arises from these populations having contextual input parameters of different variance (Fig. 5B left), and sensory input and connectivity $\mathbf{n}$ parameters characterized by opposite-sign correlations (Fig. 5B right).

In sum, the low-rank framework is able to describe the mechanism that trained networks use to solve the task – in this case, gain modulation implemented by two populations of neurons with different input and connectivity statistics. Importantly, this framework links circuit structure with neural dynamics, and neural activity with cognitive variables, enabling precise predictions that can be experimentally tested. For instance, in the rank-one RNN described here, predictions could be made about how perturbing activity (e.g., through optogenetic activation or inhibition) in one of both populations would disrupt network dynamics and behavioral performance (Dubreuil et al. 2022, Barbosa et al. 2023).

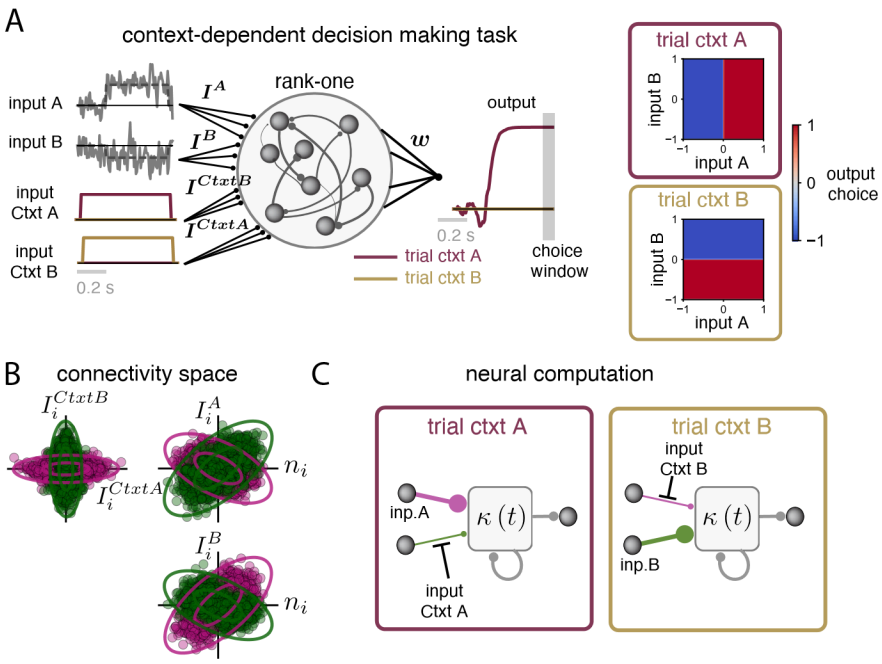

Figure 5: Low-rank recurrent network model for a context-dependent decision making task.

A Task, based on Mante et al. [2013]. Input to the network is provided along four different input patterns: $\mathbf{I}^A$ and $\mathbf{I}^B$ relay noisy sensory variables A and B (top rows), while $\mathbf{I}^{ctxtA}$ and $\mathbf{I}^{ctxtB}$ relay context variables (bottom rows). At the end of the trial (choice window, grey), the network output must reach either state +1 or state -1, depending on the sign of the relevant sensory stimulus, which is instructed by the context variable (right).

B The task could be solved by an RNN with rank-one connectivity (adapted from Dubreuil et al. [2022]). Representation of optimized parameters within the connectivity space (only a subset of input and connectivity parameters is shown). This distribution can be effectively approximated by a mixture of two Gaussians (green and purple; ellipses indicate one and three standard deviations). Accordingly, neurons are clustered into two different populations. Note that the two populations are characterized by contextual input parameters of different variance (left column), and opposite-sign correlations between the parameters of the connectivity pattern **n** and the sensory inputs.

C Schematic of the effective temporal evolution governing the latent variable, derived from mean-field theory. Arrows denote dependencies between variables, with their thickness indicating the effective strength of interactions. In context A trials (left), the context signal A acts by reducing the effective gain of the sensory input B, allowing only sensory input A to be integrated by the latent variable κ. The reverse occurs in context B trials (right).

Flexible interval timing task

In decision-making tasks, animals are generally trained to report their choice by performing an action (i.e. an eye saccade towards a specific location). However, in other cognitive tasks, the correct performance is not determined by the specific action taken but instead by the timing of that action. One such example is the interval production task based on the Cue-Set-Go task from Wang et al. [2018], in which animals are trained to perform an action (a saccade or a button press) after a time interval of instructed duration. In each trial, a “Cue” input variable indicates the duration of the time interval to be produced (Fig. 6A, bottom left). At a random time within the trial, a brief “Set” input is presented, initiating the time interval (Fig. 6A, top left). The time interval concludes when an action is taken. In an RNN setting, the desired output signal is modeled as a ramp reaching a “Go” state by the end of the

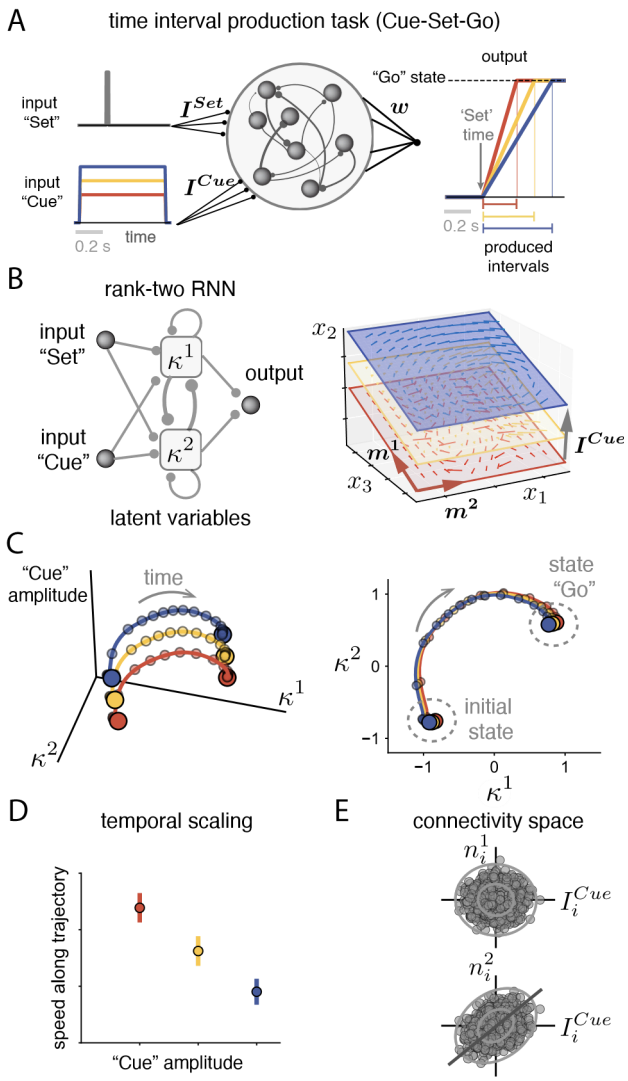

Figure 6: Low-rank recurrent network model for a flexible timing task.

A Cue-Set-Go task, based on Wang et al. [2018]. The network receives two input stimuli (left). The "Set" input is a brief pulse marking the beginning of the time interval, while the "Cue" input is persistent and indicates the duration of the interval. Different colors represent trials with varying cue amplitudes. The desired output signal (right) ramps up starting from an initial value at the time of "Set" until reaching the "Go" state at the end of the time interval.

B Rank-two implementation of the task (adapted from Beiran et al. [2023]). Left: schematic of the effective dynamics of the latent variables. The time evolution of the two latent variables is coupled. Both latents are driven by the "Set" and "Go" inputs, and contribute to the network output. Right: neural space of the RNN. Each axis corresponds to the activation of one neuron. Activity mostly evolves within the subspace spanned by connectivity patterns $\mathbf{m}^1$ and $\mathbf{m}^2$, together with the "Cue" input pattern. Different amplitudes of the "Cue" input generate different dynamics along the plane of connectivity patterns.

C Left: Activation projected on the three-dimensional space highlighted in B. Different colors correspond to trials with different cue inputs, and different produced time intervals. Right: activation projected on the plane spanned by connectivity patterns $\mathbf{m}^1$ and $\mathbf{m}^2$. The trajectories for all cue amplitudes evolve along an invariant curved manifold, from the initial state towards the "Go" state.

D Speed of neural trajectories as a function of the amplitude of the "Cue" input. Larger (resp. smaller) cues generate longer (resp. shorter) time intervals.

E Representation of optimized parameters within the connectivity space (only a subset of input and connectivity parameters is shown). Ellipses indicate one and three standard deviations of the Gaussian distribution that best approximates the optimized parameters. Importantly, parameters of the "Cue" input pattern are correlated with parameters from one of the connectivity patterns (bottom).

time interval; trials with different cues therefore correspond to ramping signals of different slopes (Fig. 6A right).

Beiran et al. [2023] trained low-rank networks to solve this task. They found that rank two was the minimal rank required for successful optimization. Similar to the previous task (Fig. 5C), the rank-two implementation could be reduced to the dynamics of two latent variables κ^1 and κ^2, in addition to the input variables (Fig. 6B left). According to the low-rank network theory, the latent variables parameterize the activation variable $\mathbf{x}$ within the plane defined by connectivity patterns $\mathbf{m}^1$ and $\mathbf{m}^2$ (Fig. 6B right, red). The activation also spans a third direction, corresponding to the cue input (Fig. 6B right, grey), which varies across trials. This results in neural trajectories that are approximately three-dimensional[9] (Fig. 6C).

One of the task requirements is that the output signal maintains the same shape across trials: for all cue amplitudes, the desired output is an identical linear ramp (Fig. 6A right), with only the slope varying. The latent variables of the optimized network satisfy this requirement by generating trajectories whose geometry is cue-independent: when neural trajectories are projected onto the plane spanned by connectivity patterns (Fig. 6C right), the activity in all trials evolves from a common initial state – corresponding to the state right before the "Set" pulse – towards the "Go" state along overlapping curved manifolds. In other words, every time the network produces a time interval, it results in an invariant activation trajectory within the $\mathbf{m}^1$−$\mathbf{m}^2$ plane.

The second task requirement is that the output signal evolves at different speeds, producing ramps with different slopes. Beiran et al. [2023] found that the speed of neural trajectories on the $\mathbf{m}^1$ −$\mathbf{m}^2$ plane is controlled by the third dimension of the activation: the "Cue" input (Fig. 6 D). Accordingly, cue inputs corresponding to long time intervals (blue) produce slower speeds compared to cue inputs for short intervals (red).

At the level of the network structure, trial-invariant latent variable dynamics combined with flexible speeds is implemented through statistical dependencies between the trained parameters of the "Cue" input and the connectivity patterns $\mathbf{n}^1$ and $\mathbf{n}^2$. In particular, the optimized networks displayed strong correlations between the parameters of the "Cue" input pattern and at least one of the connectivity patterns, $\mathbf{n}^1$ or $\mathbf{n}^2$. In the network shown in Fig. 6E, the cue pattern is correlated with $\mathbf{n}^2$. Removing this correlation from trained parameters impairs the network's ability to modulate speed.

Finally, the detailed understanding of this rank-two implementation generated testable predictions on the relationship between neural activity and behavior. In the RNN model, exciting or inhibiting neurons tuned to the cue input selectively affects the speed of neural trajectories and therefore the produced time interval, without disrupting the overall sequence of neural states [Beiran et al., 2023]. Altogether, this second example task illustrates how the framework of low-rank RNNs can be useful to reverse-engineer task implementations that are characterized by a strong dynamical component, by establishing a direct link between the dynamical properties of neural trajectories and the network connectivity.

[9] The "Set" input elicits a fast transient response that adds one additional dimension to the subspace spanned by the activation; for simplicity, that component is not illustrated.

Low-rank models for synaptic plasticity and learning

In all examples discussed in the previous section, low-rank RNNs have been used to model neural activity and behavior in animals that are proficient and demonstrate high performance levels in the task. There is however, at the experimental level, increasing interest in studying the learning processes that enable animals to reach proficiency over the course of behavioral training [Khan et al., 2018, Reinert et al., 2021, Sun et al., 2022]. This growing focus has been driven by advancements in recording techniques, which enable researchers to track neural activity over multiple days using both single-cell resolution electrophysiology and imaging [Steinmetz and al, 2021, Driscoll et al., 2017]. Motivated by this line of experimental work, new algorithms have been proposed to optimize low-rank RNNs that are plastic, and therefore can capture the changes in synaptic connectivity and dynamics that occur during learning [Pellegrino et al., 2023].

While awaiting the application of these approaches to experimental data, low-rank RNNs have been effectively used to study a different type of learning: the *artificial* learning process that RNNs undergo while being optimized on a task. Studying learning in artificial neural networks is relevant to the field for several reasons. First, understanding how large, distributed systems of simple units can coordinate to update their internal structure in order to fulfill a specific task provides valuable insights into the broad topic of learning and adaptation in biological networks, of which brain networks are just one particular example. Second, grasping the details of the optimization process, along with its inherent biases, is essential for better characterizing the space of possible network solutions, and potentially identifying classes of models that more closely align with experimental data [Turner et al., 2021].

Early optimization algorithms developed for RNNs in the context of neuroscience involved updates to synaptic connectivity that had an explicit low-rank constraint, corresponding to interactions within the circuit to pass through a low-dimensional bottleneck [Jaeger, 2001, Sussillo and Abbott, 2009, Eliasmith and Anderson, 2004] (Fig. 1G-H). For these algorithms, the mathematical framework of low-rank RNNs serves as a natural and valuable analytical tool. This framework has been used to investigate, for example, the relationship between the dimensionality of neural activity and learning proficiency [Susman et al., 2021], the dynamical stability of learned activity trajectories [Rivkind and Barak, 2017, Barak and Romani, 2021], and the emergence of spurious multi-stable attractors during the formation of memories [Mastrogiuseppe and Ostojic, 2019]. Understanding learning in low-rank networks is also of specific relevance to biology, given that the brain's learning system – which involves both cortical and subcortical areas (e.g., thalamus, basal ganglia) – is also organized around bottleneck structures [Lakshminarasimhan et al., 2024].

Low-rank RNNs can, however, also be valuable in the study of algorithms that do not impose low dimensional updates. These include the foundational algorithm of modern machine intelligence, gradient descent based on backpropagation [LeCun et al., 2015], and its RNN-specific formulation known as backpropagation through time [Marschall et al., 2020]. A significant body of research in the machine learning literature has put forward the hypothesis that gradient-based algorithms may be biased towards converging to synaptic connectivity matrices of low rank [Hu et al., 2023]. In the context of systems neuroscience, it has been observed that RNNs trained on behavioral tasks converge to solutions where connectivity updates are low-rank [Schuessler et al., 2020b]. For these solutions, the low-rank mathematical formalism has proven useful for understanding how task-related structures emerge in connectivity from initially unstructured synapses, and revealed that synergies between the learned and the initial connectivity play an important role in speeding up learning [Schuessler et al., 2020b]. The idea that, although task unrelated, initial connectivity may play a

fundamental role in learning has also emerged in other lines of study [Mastrogiuseppe et al., 2023, Susman et al., 2021, Schuessler et al., 2024, Jacot et al., 2018, Liu et al., 2024].

A fundamental question in artificial neural networks is that of multi-tasking and continual learning – namely, how a single network can effectively learn and perform multiple tasks while retaining proficiency across them. The low-rank formalism enables the design of compact task solutions, for which task-related neural activity is confined to specific low-dimensional subspaces. Building on this, multi-task RNNs have been constructed as the addition of low-rank matrices, each associated with distinct tasks and subspaces, into the synaptic connectivity [Dubreuil et al., 2022, Duncker et al., 2020]. While promising in some scenarios, this approach still presents challenges [Chiappa et al., 2024] and open theoretical questions. First, different computations may need to operate on shared neural subspaces (for instance, when tasks share similar input or output dimensions or depend on similar internal dynamics [Driscoll et al., 2024]). In mathematical terms, this situation corresponds to connectivities (Eq. 2) where the connectivity patterns associated with different tasks are not orthogonal. Mean-field theory indicates that this scenario can potentially lead to catastrophic forgetting, as learning a new task via a new low-rank term profoundly reshapes neural activity in previously learned tasks. Less intuitively, even in cases where connectivity patterns are orthogonal, interference issues can arise. This is because adding up orthogonal low-rank connectivities can alter the dynamical stability of previously learned activity trajectories [Beiran et al., 2021]. Thus, while the low-rank formalism offers a structured approach to multi-task learning, addressing these interference and stability challenges is essential for advancing continual learning in neural networks.

Further considerations and discussion

This article provides an overview of low-rank recurrent neural networks, a specific class of mathematical models for the study of brain activity at single-cell resolution during behavioral tasks. These models assume that connectivity in the underlying circuit is approximately low-dimensional, meaning it can be reconstructed from a small number of task relevant directions. Mathematically, this translates into a connectivity matrix that is approximately low rank. This assumption makes the models mathematically tractable and therefore well-suited to investigate the link between structure and function in large-scale, non-linear brain circuits.

A key feature of these models is that the activity they generate is low-dimensional. In experimental research, dimensionality reduction algorithms are widely used to interpret data from large-scale recordings during behavioral tasks, which often exhibit low dimensionality [Churchland et al., 2012, Mante et al., 2013, Wang et al., 2018]. Consequently, low-rank RNNs provide a natural mathematical framework for developing theoretical hypotheses about these experiments [Dubreuil et al., 2022, Beiran et al., 2023]. Another context in which low dimensionality has proven valuable for interpreting experimental datasets is the study of interactions among populations of neurons from different brain areas. If activity within a single area can be captured by a small number of latent variables (low-dimensional), the activity co-fluctuations representing interactions between areas is even lower dimensional [Semedo et al., 2019, Ebrahimi et al., 2022]. This insight has inspired theoretical studies [Barbosa et al., 2023, Clark and Beiran, 2024], which model the difference in dimensionality between within- and across-area activity by employing a difference in connectivity rank within and across areas.

It is important to emphasize that low-rank connectivity is not the only mechanism capable of generating low-dimensional activity in RNNs. One possible alternative involves low-dimensional input drives derived from sensory stimuli or activity in other brain areas. When activity of different neurons

synchronizes with a common temporally varying signal, the resulting dimensionality is typically low. This type of mechanism is supported by experimental evidence showing that neural activity in a given brain region can be locked to signals generated elsewhere [Sauerbrei et al., 2019, van Kempen et al., 2021].

Low-rank RNNs were initially developed with the primary focus of explaining how large-scale networks of neurons give rise to cognitive functions. Being mostly used as conceptual models for distributed computation, these models often abstract away from biological realism, and do not incorporate detailed biological constraints on their dynamics and synaptic connectivity. Integrating biological constraints into low-rank RNNs remains an important challenge in ongoing research. One key biological constraint is Dale's principle, which states that the sign of a synaptic connection (positive vs negative) depends solely on the identity of the pre-synaptic neuron (excitatory, E, vs inhibitory, I). This E-I structure leads to specific signatures in neural activity, such as damped or amplified responses to transiently presented inputs [Murphy and Miller, 2009]. At first glance, E-I and low-rank connectivity structures might appear fundamentally different. E-I structure is expressed locally, imposing constraints on the synaptic weights based on the pre-synaptic neuron's identity. In contrast, the rank is a global property of the connectivity matrix, reflecting the overall structure of synaptic connections and shaping emergent activity at the population level (Fig. 2B). However, these two types of structure can be reconciled, as local constraints like Dale's law can effectively be described by low-rank networks with multiple populations [Shao and Ostojic, 2023]. Moreover, the low-rank framework can be extended to take into account other local connectivity features, such as the over-representation of reciprocal connectivity motifs [Shao and Ostojic, 2023, Shao et al., 2025]. Additionally, progress has been made in extending the low-rank framework to incorporate other biologically relevant connectivity features, such as sparsity [Herbert and Ostojic, 2022].

Another important difference between RNNs and biological circuits is that in most low-rank models, the non-linear function governing single-neurons activation is bounded, which ensures the stability of activity by limiting firing rates. Biological neurons, instead, operate within input ranges far from any saturating bounds [Priebe et al., 2004], and maintain stability dynamically through inhibitory interactions. Further research is needed to determine whether and how the dynamical mechanisms identified in bounded low-rank RNNs extend to unbounded models. Finally, while the low-rank framework discussed here focuses on RNN models with rate-based dynamics, neurons communicate with each other via spikes. Interestingly, multiple studies indicate that low-rank connectivity can also simplify the design and analysis of RNNs with spiking dynamics [Cimesa et al., 2023, Mancoo et al., 2020, Podlaski and Machens, 2024]. In particular, it has been shown that neurons interacting through spikes can still generate low-dimensional activity at the population level that follow continuous rate-like dynamics [DePasquale et al. 2023, Schmutz et al. 2025].

The section "Low-rank models of cognitive tasks" discussed how RNN models with low-rank connectivity can be optimized to implement behavioral tasks. However, low-rank RNNs can also be trained to replicate activity patterns from experimental datasets [Valente et al., 2022a], following a strategy that has been widely used in the context of RNNs with generic connectivity [Sussillo et al. 2015, Rajan et al. 2016]. In this approach, each unit in the RNN is matched to an experimentally recorded neuron, with spiking activity binned, smoothed or trial-averaged to produce a target signal resembling firing rates. The fitting technique proposed in Valente et al. [2022b] involves brute-force optimization of the low-rank model parameters, similar to methods used for task learning. The low-rank connectivity patterns are optimized in parallel with the input patterns, while the time-varying input signals that drive the RNN are fixed according to user-specified assumptions. The connectivity matrix rank serves as a hyperparameter that can be selected, for example, through cross-validation. Because

the connectivity rank directly influences activity dimensionality, low-rank RNNs are expected to provide accurate and parsimonious fits to experimental data primarily in cases where the activity dimensionality is relatively low.

Data-trained low-rank networks are closely related to state-space models, a class of statistical models designed to extract low-dimensional patterns and dynamics from data. Many state-space approaches have been developed specifically for neuroscience applications [Zhao and Park, 2017, Duncker et al., 2019, Genkin et al., 2022, Langdon and Engel, 2025]. Among the simplest, state-space linear dynamical systems (LDSs) [Ghahramani and Hinton, 1996, Wei et al., 2019] assume that recorded neural activity is a linear projection, corrupted with noise, of a small number of latent variables, a structure similar to that seen in low-rank RNNs (see Eq. 5). In general, state-space models and low-rank RNNs share a number of common features – above all, the core assumption that temporal dynamics in large neuronal populations can be reduced to the dynamics of a small number of latent variables, which are reflected into the neural activity according to simple geometrical relationships. However, the two approaches are also significantly different from each other. In contrast to low-rank RNNs, state-space models return a statistically richer portrait of the data by providing a quantitative description not only of the latent dynamics, but also of the inherent noise in the latent and the observation models. On the contrary, the focus of low-rank RNNs is on model simplicity, and mathematical insight. The mathematical formalism behind the two classes of models also shows important differences. For instance, state-space models often rely on discrete-time updates and probabilistic formulations of the activity, while low-rank RNNs rely on continuous-time dynamics as expressed by deterministic ordinary differential equations. Mathematically, an exact mapping between state-space models and low-rank RNNs has not been fully traced yet. When low-rank RNNs use a linear activation function, a close relationship (although not a full equivalence) exists between the two models [Valente et al., 2022a]. Recently, a statistical approach aimed at reconciling these two classes of models in the context of non-linear dynamics has been proposed [Pals et al., 2024]. This work considers low-rank RNNs in the presence of stochastic inputs, and relies on a probabilistic formulation together with a variational approximation to fit these models to the data.

In summary, low-rank RNNs provide a conceptual framework for computation in neural circuits that balances expressiveness, mathematical tractability, and biological plausibility. By constraining the synaptic connectivity to have low rank, they can capture the low-dimensional neural activity observed in cognitive tasks, while establishing a formal link between network connectivity, neural population dynamics, and computation. Low-rank RNNs serve two complementary roles. First, as a subject of theoretical study, they provide parsimonious models of distributed neural computation that can target the same questions that neuroscientists explore in the brain (for instance: task performance, generalization and learning, interference across tasks, and the associated circuit structure). Second, they serve as a data-driven tool for neural analyses, generating hypotheses of the latent dynamics and the underlying biological structure of the recorded signals. As the fields of cognitive and neural sciences advances with novel tasks and larger scale recordings, low-rank RNNs offer an invaluable conceptual tool for understanding how neural circuits give rise to such diverse cognitive functions.

Acknowledgments

M.B. is supported by NIH award R01EB029858, the Kavli Foundation, and the Gatsby Charitable Foundation GAT3708. F.M. is supported by the Transition To Independence grant 00002313 from Simons Foundation. The authors would like to thank Srdjan Ostojic for useful feedback on the manuscript.

References

E. Aksay, G. Gamkrelidze, H. Seung, R. Baker, and D. Tank. In vivo intracellular recording and perturbation of persistent activity in a neural integrator. *Nat. Neurosci.*, 4:184–193, 2001.

D. Aronov, R. Nevers, and D. Tank. Mapping of a non-spatial dimension by the hippocampal–entorhinal circuit. *Nature*, 543:719–722, 2017.

O. Barak. Recurrent neural networks as versatile tools of neuroscience research. *Curr. Opin. Neurobiol.*, 46: 1–6, 2017.

O. Barak and S. Romani. Mapping low-dimensional dynamics to high-dimensional neural activity: A derivation of the ring model from the neural engineering framework. *Neural Comput.*, 33:827–852, 2021.

J. Barbosa, R. Proville, C. Rodgers, M. DeWeese, S. Ostojic, and Y. Boubenec. Early selection of task-relevant features through population gating. *Nature Communications*, 14:6837, 2023.

M. Beiran, A. Dubreuil, A. Valente, F. Mastrogiuseppe, and S. Ostojic. Shaping dynamics with multiple populations in low-rank recurrent networks. *Neural Comput.*, 33(6):1572–1615, 2021.

M. Beiran, N. Meirhaeghe, H. Sohn, M. Jazayeri, and S. Ostojic. Parametric control of flexible timing through low-dimensional neural manifolds. *Neuron*, 111(5):739–753, 2023.

R. Ben-Yishai, R L. Bar-Or, and H. Sompolinsky. Theory of orientation tuning in visual cortex. *Proc. Natl. Acad. Sci. USA*, 92(9):3844–3848, 1995.

M. Boerlin, C. K. Machens, and S. Deneve. Predictive coding of dynamical variables in balanced spiking networks. *PLOS Comput. Biol.*, 9(11):1–16, 11 2013.

G. Bondanelli and S. Ostojic. Coding with transient trajectories in recurrent neural networks. *PLOS Comput Biol*, 16(2):e1007655, 2020.

S. Brincat, M. Siegel, C. von Nicolai, and E. Miller. Gradual progression from sensory to task-related processing in cerebral cortex. *Proc. Natl. Acad. Sci. USA*, 115(30):e7202–e7211, 2018.

N. Brunel. Dynamics of sparsely connected networks of excitatory and inhibitory spiking neurons. *J. Comput. Neurosci.*, 8(3):183–208, 2000.

Y. Burak and I. R. Fiete. Accurate path integration in continuous attractor network models of grid cells. *PLOS Computat. Biol.*, 5:1–16, 2009.

F. Carnevale, V. de Lafuente, R. Romo, O. Barak, and N. Parga. Dynamic Control of Response Criterion in Premotor Cortex during Perceptual Detection under Temporal Uncertainty. *Neuron*, 86(4):1067–1077, 2016.

W. Chaisangmongkon, S. Swaminathan, D. Freedman, and X.-J. Wang. Computing by robust transience: How the fronto-parietal network performs sequential, category-based decisions. *Neuron*, 93(6):1504–1517.e4, 2017.

A. Chiappa, P. Tano, N. Patel, A. Ingster, A. Pouget, and A. Mathis. Acquiring musculoskeletal skills with curriculum-based reinforcement learning. *Neuron*, 112(23):3969–3983.e5, 2024.

M. Churchland and K. Shenoy. Temporal complexity and heterogeneity of single-neuron activity in premotor and motor cortex. *J. Neurophysiol.*, 97(6):4235–4257, 2007.

M. Churchland, J.Cunningham, M. Kaufman, J. Foster, P. Nuyujukian, S. Ryu, and K. Shenoy. Neural population dynamics during reaching. *Nature*, 487(7405):51–56, 2012.

L. Cimesa, L. Ciric, and S. Ostojic. Geometry of population activity in spiking networks with low-rankstructure. *PLOS Comput. Biol.*, 19(8):1–34, 2023.

D. Clark and M. Beiran. Structure of activity in multiregion recurrent neural networks. *Proc. Natl. Acad. Sci. USA*, 112(10):e2404039122, 2024.

A. Compte, N. Brunel, P. Goldman-Rakic, and X.-J. Wang. Synaptic mechanisms and network dynamics underlying spatial working memory in a cortical network model. *Cereb. Cortex*, 10:910, 2000.

C. Cueva, A. Saez, E. Marcos, A. Genovesio, M. Jazayeri, R. Romo, D. Salzman, M. Shadlen, and S. Fusi. Low-dimensional dynamics for working memory and time encoding. *Proc. Natl. Acad. Sci. USA*, 117(37): 23021–23032, 2020.

J. P. Cunningham and B. M. Yu. Dimensionality reduction for large-scale neural recordings. *Nat. Neurosci.*, 17(11):1500–1509, 2014.

A. Das and I. Fiete. Systematic errors in connectivity inferred from activity in strongly recurrent networks. *Nat. Neurosci.*, 23(10):1286–1296, 2020.

P. Dayan and L. Abbott. *Theoretical Neuroscience. Computational and Mathematical Modeling of Neural Systems.* MIT press, 2001.

B. DePasquale, D. Sussillo, L. Abbott, and M. Churchland. The centrality of population-level factors to network computation is demonstrated by a versatile approach for training spiking networks. *Neuron*, 111 (5):631–649.e10, 2023.

L. Driscoll, N. Pettit, M. Minderer, S. Chettih, and C. Harvey. Dynamic reorganization of neuronal activity patterns in parietal cortex. *Cell*, 170:986–999, 2017.

L. Driscoll, K. Shenoy, and D. Sussillo. Flexible multitask computation in recurrent networks utilizes shared dynamical motifs. *Nat. Neurosci.*, 27 :1349–1363, 2024.

A. Dubreuil, A. Valente, M. Beiran, F. Mastrogiuseppe, and S. Ostojic. The role of population structure in computations through neural dynamics. *Nat. Neurosci.*, 25:783–794, 2022.

L. Duncker, G. Bohner, J. Boussard, and M. Sahani. Learning interpretable continuous-time models of latent stochastic dynamical systems. *ICLR*, 2019.

L. Duncker, L. Driscoll, K. Shenoy, M. Sahani, and D. Sussillo. Organizing recurrent network dynamics by task-computation to enable continual learning. *Adv. Neural Inf. Process. Syst.*, 33, 2020.

S. Ebrahimi, J. Lecow, O. Rumyantsev, T. Tasci, Y. Zhang, C. Irimia, J. Li, S. Ganguli, and M. Schnitzer. Emergent reliability in sensory cortical coding and inter-area communication. *Nature*, 605:713–721, 2022.

C. Eliasmith and A. Anderson. *Neural Engineering - Computation, Representation, and Dynamics in Neurobiological Systems*. MIT press, 2004

D. Freedman, M. Riesenhuber, T. Poggio, and E. Miller. Categorical representation of visual stimuli in the primate prefrontal cortex. *Science*, 291(5502):312–316, 2001.

P. Gao and S. Ganguli. On simplicity and complexity in the brave new world of large-scale neuroscience. *Curr. Opin. Neurobiol.*, 32:148–55, 2015.

M. Genkin, O. Hughes, and T. Engel. Learning non-stationary langevin dynamics from stochastic observations of latent trajectories. *Nat. Commun.*, 12:5986, 2022.

Z. Ghahramani and G. Hinton. Parameter estimation for linear dynamical systems. *Technical Report CRGTR-96-2*, 1996.

V. L. Girko. Circular law. *Theory Probab. Appl.*, 29(4):694–706, 1985.

J. Goldberg, U. Rokni, and H. Sompolinsky. Patterns of ongoing activity and the functional architecture of the primary visual cortex. *Neuron*, 42(3):489–500, 2004.

E. Herbert and S. Ostojic. The impact of sparsity in low-rank recurrent neural networks. *PLOS Comput Biol*, 18(8):1–21, 2022.

J. J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proc. Natl. Acad. Sci. USA*, 79(8):2554–2558, 1982.

M. Hu, H. Mobahi, R. Zhang, B. Cheung, P. Agrawal, and P. Isola. The low-rank simplicity bias in deep networks. *Trans. Machine Learn. Res.*, 3, 2023.

A. Jacot, F. Gabriel, and C. Hongler. Neural tangent kernel: Convergence and generalization in neural networks. *Adv. Neural Inf. Process. Syst.*, 31:8571–8580, 2018.

H. Jaeger. The "echo state" approach to analysing and training recurrent neural networks - with an erratum note. *GMD Report*, 2001.

M. Jazayeri and S. Ostojic. Interpreting neural computations by examining intrinsic and embedding dimensionality of neural activity. *Curr. Opin. Neurobiol.*, 70:113–120, 2021.

K. Kay, N. Biderman, R. Khajeh, M. Beiran, C. Cueva, D. Shohamy, G. Jensen, X. Wei, V. Ferrera and L. Abbott. Emergent neural dynamics and geometry for generalization in a transitive inference task. *PLOS Comput. Biol.*, 20(4), e1011954, 2024.

T. Kao, M. Sadabadi, and G. Hennequin. Optimal anticipatory control as a theory of motor preparation: A thalamo-cortical circuit model. *Neuron*, 109(9):1567–1581.e12, 2021.

A. Khan, J. Poort, A. Chadwick, A. Blot, M. Sahani, T. Mrsic-Flogel, and S. Hofer. Distinct learning induced changes in stimulus selectivity and interactions of gabaergic interneuron classes in visual cortex. *Nat. Neurosci.*, 21(6):851–859, 2018.

D. Kingma and J. Ba. Adam: A method for stochastic optimization. *ICLR*, 2015.

G. La Camera. *The Mean Field Approach for Populations of Spiking Neurons*, pages 125–157. Springer, 2022.

K. Lakshminarasimhan, M. Xie, J. Cohen, B. Sauerbrei, A. Hantman, A. Litwin-Kumar, and S. Escola. Specific connectivity optimizes learning in thalamocortical loops. *Curr. Biol.*, 43:114059, 2024.

I. Landau and H. Sompolinsky. Coherent chaos in a recurrent neural network with structured connectivity. *PLOS Comput. Biol.*, 14(12):1–27, 12 2018.

C. Langdon and T. Engel. Latent circuit inference from heterogeneous neural responses during cognitive tasks. *Nat. Neurosci.*, 28, 665–675, 2025.

F. Lanore, A. Cayco-Gajic, H. Gurnani, D. Coyle, and A. Silver. Cerebellar granule cell axons support high-dimensional representations. *Nat. Neurosci.*, 24:1142–1150, 2021.

Y. LeCun, Y. Bengio, and G. Hinton. Deep learning. *Nature*, 521:436, 2015.

A. Litwin-Kumar, K. Harris, R. Axel, H. Sompolinsky, and L. Abbott. Optimal degrees of synaptic connectivity. *Neuron*, 93(5):1153 – 1164.e7, 2017.

Y. Liu, A. Baratin, J. Cornford, S. Mihalas, E. SheaBrown, and G. Lajoie. How connectivity structure shapes rich and lazy learning in neural circuits. *ICLR*, 2024.

L. Logiaco, L. Abbott, and S. Escola. Thalamic control of cortical dynamics in a model of flexible motor sequencing. *Cell Rep.*, 35(9):109090, 2021.

A. Mancoo, S. Keemink, and C. Machens. Understanding spiking networks through convex optimization. *Adv. Neural Inf. Process. Syst.*, 34, 2020.

J. Manley, S. Lu, K. Barber, J. Demas, H. Kim, D. Meyer, F. Martinez Traub, and A. Vaziri. Simultaneous, cortex-wide dynamics of up to 1 million neurons reveal unbounded scaling of dimensionality with neuron number. *Neuron*, 112(20):1694–1709.e5, 2024.

V. Mante, D. Sussillo, K. V. Shenoy, and W. T. Newsome. Context-dependent computation by recurrent dynamics in prefrontal cortex. *Nature*, 503(7474):78–84, 2013.

O. Marschall, K. Cho, and C. Savin. A unified framework of online learning algorithms for training recurrent neural networks. *J. Mach. Learn. Res.*, 21(1), 2020.

F. Mastrogiuseppe and S. Ostojic. Linking connectivity, dynamics and computations in low-rank recurrent neural networks. *Neuron*, 99(3):609–623, 2018.

F. Mastrogiuseppe and S. Ostojic. A geometrical analysis of global stability in trained feedback networks. *Neural Computat.*, 31(6):1139–1182, 2019.

F. Mastrogiuseppe, N. Hiratani, and P. Latham. Evolution of neural activity in circuits bridging sensory and abstract knowledge. *eLife*, 12:e79908., 2023.

H. Moritz and D. Dahmen. *Statistical Field Theory for Neural Networks*. Springer, 2020.

B. Murphy and K. Miller. Balanced amplification: A new mechanism of selective amplification of neural activity patterns. *Neuron*, 61:635–648, 2009.

A. Nair, T. Karigo, B. Yang, S. Ganguli, M. Schnitzer, S. Linderman, S. Anderson, and A. Kennedy. An approximate line attractor in the hypothalamus encodes an aggressive state. *Cell*, 186:178–193, 2023.

M. Pals, E. Sagtekin, F. Pei, M. Gloeckler, and J. Macke. Inferring stochastic low-rank recurrent neural networks from neural data. *Adv. Neural Inf. Process. Syst.*, 37, 2024.

R. Pascanu, T. Mikolov, and Y. Bengio. On the difficulty of training recurrent neural networks. *ICML*, pages 1310–1318, 2013.

A. Pellegrino, A. Cayco Gajic, and A. Chadwick. Low tensor rank learning of neural dynamics. *Adv. Neural Inf. Process. Syst.*, 36, 2023.

U. Pereira and N. Brunel. Attractor dynamics in networks with learning rules inferred from in vivo data. *Neuron*, 99(1):227–238.e4, 2018.

W. Podlaski and C. Machens. Approximating nonlinear functions with latent boundaries in low-rank excitatory-inhibitory spiking networks. *Neural Comput*, 36:803–857, 2024.

N. Priebe, F. Mechler, M. Carandini, and D. Ferster. The contribution of spike threshold to the dichotomy of cortical simple and complex cells. *Nat. Neurosci.*, 7(10):1113–1122, 2004.

K. Rajan and L. F. Abbott. Eigenvalue spectra of random matrices for neural networks. *Phys. Rev. Lett.*, 97:188104, 2006.

K. Rajan, C. Harvey and D. Tank. Recurrent Network Models of Sequence Generation and Memory. *Neuron*, 90(1):128–142, 2006.

D. Redish, A. Elga, and D. Touretzky. A coupled attractor model of the rodent head direction system. *Netw. Comput. Neural Syst.*, 7(4):671, 1996.

S. Reinert, M. Hubener, T. Bonhoeffer, and P. Goltstein. Mouse prefrontal cortex represents learned rules for categorization. *Nature*, 593(7859):411–417, 2021.

A. Renart, J. de la Rocha, P. Bartho, L. Hollender, N. Parga, A. Reyes, and K. D. Harris. The asynchronous state in cortical circuits. *Science*, 327(5965):587–590, 2010.

A. Rivkind and O. Barak. Local dynamics in trained recurrent neural networks. *Phys. Rev. Lett.*, 118: 258101, 2017.

J. Roitman and M. Shadlen. Response of neurons in the lateral intraparietal area during a combined visual discrimination reaction time task. *J. Neurosci.*, 22(21):9475–9489, 2002.

R. Romo, C. Brody, A. Hernandez, and L. Lemus. Neuronal correlates of parametric working memory in the prefrontal cortex. *Nature*, 399:470–473, 1999.

B. Sauerbrei, J.-Z. Guo, J. Cohen, M. Mischiati, W. Guo, M. Kabra, N. Verma, B. Mensh, K. Branson, and A. Hantman. Cortical pattern generation during dexterous movement is input-driven. *Nature*, 577: 386–391, 2019.

R. Schaeffer, M. Khona, L. Meshulam, International Brain Laboratory, and I. Fiete. Reverse-engineering recurrent neural network solutions to a hierarchical inference task for mice. *Adv. Neural Inf. Process. Syst.*, 33, 2020.

V. Schmutz,, J. Brea, and Gerstner, W. Emergent rate-based dynamics in duplicate-free populations of spiking neurons. *Phys. Rev. Lett.*, 134(1), 018401, 2025.

F. Schuessler, A. Dubreuil, F. Mastrogiuseppe, S. Ostojic, and O. Barak. Dynamics of random recurrent networks with correlated low-rank structure. *Phys. Rev. Research*, 2:013111, 2020a.

F. Schuessler, F. Mastrogiuseppe, A. Dubreuil, S. Ostojic, and O. Barak. The interplay between randomness and structure during learning in RNNs. *Adv. Neural Inf. Process. Syst.*, 33:13352–13362, 2020b.

F. Schuessler, F. Mastrogiuseppe, S. Ostojic, and O. Barak. Aligned and oblique dynamics in recurrent neural networks. *eLife*, 13:RP93060, 2024.

J. Semedo, A. Zandvakili, C. Machens, B. Yu, and A. Kohn. Cortical areas interact through a communication subspace. *Neuron*, 102(1):249–259, 2019.

H. Seung. How the brain keeps the eyes still. *Proc. Natl. Acad. Sci. USA*, 93(23):13339–13344, 1996.

M. N. Shadlen and W. T. Newsome. The variable discharge of cortical neurons: Implications for connectivity, computation, and information coding. *J. Neurosci.*, 18(10):3870–3896, 1998.

Y. Shao, D. Dahmen, S. Recanatesi, E. Shea-Brown and S. Ostojic. Impact of Local Connectivity Patterns on Excitatory-Inhibitory Network Dynamics. *Phys. Rev. X Life*, 3(2):023008, 2025.

H. Sompolinsky, A. Crisanti, and H. J. Sommers. Chaos in random neural networks. *Phys. Rev. Lett.*, 61: 259–262, 1988.

N. Steinmetz and al. Neuropixels 2.0: A miniaturized high-density probe for stable, long-term brain recordings. *Science*, 372:eabf4588, 2021.

C. Stringer, M. Pachitariu, N. Steinmetz, C. Reddy, M. Carandini, and K. Harris. Spontaneous behaviors drive multidimensional, brainwide activity. *Science*, 364(6437):255, 2019.

S. Strogatz. *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry and Engineering*. Westview Press, 2000.

X. Sun, D. O'Shea, M. Golub, E. Trautmann, S. Vyas, S. Ryu, and K. Shenoy. Cortical preparatory activity indexes learned motor memories. *Nature*, 602:274–279, 2022.

L. Susman, F. Mastrogiuseppe, N. Brenner, and O. Barak. Quality of internal representation shapes learning performance in feedback neural networks. *Phys. Rev. Research*, 3:013176, 2021.

D. Sussillo and L.F. Abbott. Generating coherent patterns of activity from chaotic neural networks. *Neuron*, 63(4):544 – 557, 2009.

D. Sussillo and O. Barak. Opening the black box: Low-dimensional dynamics in high-dimensional recurrent neural networks. *Neural Comput.*, 25(3):626–649, 2013.

D. Sussillo, M. Churchland, M. Kaufman, and K. Shenoy. A neural network that finds a naturalistic solution for the production of muscle activity. *Nat. Neurosci.*, 18, 1025-1033, 2015.

T. Tao. Outliers in the spectrum of iid matrices with bounded rank perturbations. *Probab. Theory Relat. Fields*, 155(1):231–263, 2013.

M. Tsodyks and T. Sejnowski. Associative memory and hippocampal place cells. *Int. J. Neural Sys.*, 6:81–86, 1995.

E. Turner, K. Dabholkar, and O. Barak. Charting and navigating the space of solutions for recurrent neural networks. *Adv. Neural Inf. Process. Syst.*, 34, 2021.

A. Valente, S. Ostojic, and J. Pillow. Probing the relationship between linear dynamical systems and low-rank recurrent neural network models. *Neural Comput.*, 34 (9):1871–1892, 2022a.

A. Valente, J. Pillow, and S. Ostojic. Extracting computational mechanisms from neural data using low-rank RNNs. *Adv. Neural Inf. Process. Syst.*, 35, 2022b.

J. van Kempen, M. Gieselmann, M. Boyd, N. Steinmetz, T. Moore, and T. Engel, and A. Thiele. Top-down coordination of local cortical state during selective attention. *Neuron*, 109:894–904, 2021.

C. van Vreeswijk and H. Sompolinsky. Chaos in neuronal networks with balanced excitatory and inhibitory activity. *Science*, 274(5293):1724–1726, 1996.

A. Vinograd, A. Nair, J. Kim, S. Linderman, and D. Anderson. Causal evidence of a line attractor encoding an affective state. *Nature*, (634):910–918, 2024.

J. Wallis, K. Anderson, and E. Miller. Single neurons in prefrontal cortex encode abstract rules. *Nature*, 411 (6840):953–956, 2001.

J. Wang, D. Narain, E. Hosseini, and M. Jazayeri. Flexible timing by temporal scaling of cortical responses. *Nat. Neurosci.*, 21(1):102–110, 2018.

X.-J. Wang. Probabilistic decision making by slow reverberation in cortical circuits. *Neuron*, 36(5):955–968, 2002.

Z. Wei, H. Inagaki, N. Li, K. Svoboda, and S. Druckmann. An orderly single-trial organization of population dynamics in premotor cortex predicts behavioral variability. *Nat. Commun.*, (10):216, 2019.

K. Zhang. Representation of spatial orientation by the intrinsic dynamics of the head-direction cell ensemble: a theory. *J. Neurosci.*, 16(6):2112–2126, 1996.

Y. Zhao and M. Park. Variational latent gaussian process for recovering single-trial dynamics from population spike trains. *Neural Comput.*, 29(5):1293–1316, 2017.